Xueren Ge

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Summary

Ph.D. candidate in **LLM post-training, multi-agent systems, and retrieval-augmented generation** on real-world clinical data. First-author papers at **ACL, AAAI (oral), and EMNLP**; LLM research intern at **Samsung Research America**. Seeking a Summer 2027 research or machine learning internship in LLM post-training and agentic systems.

Education

University of Virginia — Jun. 2022 – Dec. 2027 (expected)
Ph.D. in Computer Engineering, GPA 3.88/4.0 — *Charlottesville, VA*

Georgia Institute of Technology — Aug. 2020 – May 2022
M.S. in Electrical and Computer Engineering, GPA 3.91/4.0 — *Atlanta, GA*

Chongqing University — Sep. 2016 – Jun. 2020
B.E. in Electrical Engineering and Automation, GPA 3.58/4.0 — *Chongqing, China*

Industry Experience

Samsung Research America — May 2026 – Aug. 2026
LLM / Gen AI Research Intern | Mentor: Jilong Kuang — *Mountain View, CA*

- **FHIR-GRAMS: Graph-Normalized Records and a Multi-Agent System for Question Answering over FHIR EHRs**
  LLM agents, skills, tool use, information retrieval, Gemini API – paper under review
  - Designed a **planner–specialist–composer** framework: graph normalization flattens nested, reference-linked FHIR resources into canonical views, specialists project query-relevant evidence, and a composer reasons with retrieved skills and **deterministic tools instead of generated code**.
  - Beat single-agent CodeAct and the skill-optimized GRASP agent on **4 benchmarks** with two Gemini backbones: **+11.6 / +8.0 points** LLM-judged accuracy over the strongest baseline, best exact match on every benchmark, **100% / 96.7%** on MedAgentBench v1/v2, and **up to 2× fewer tokens** per query than multi-turn baselines.

SenseTime — Dec. 2020 – Jun. 2021
Algorithm Developer Intern | Mentor: Ruodai Li — *Shenzhen, China*

- **Contactless Vital-Sign Estimation from Thermal Video for the SenseThunder Air product**
  Python, signal processing, computer vision – 2 patents (CN113887474B, granted; WO2023061042A1)
  - Built a contactless heart-rate, respiratory-rate, and temperature estimator: landmark detection and homography track thermal regions of interest, and FFT plus bandpass filtering recover weak periodic signals from the video.
  - Delivered an end-to-end Python framework for preprocessing, evaluation, and error analysis for deployment.

Research Experience

University of Virginia, Link Lab — Jun. 2022 – Present
Graduate Research Assistant | Advisor: Homa Alemzadeh — *Charlottesville, VA*

- **FHIR-R1: A Healthcare QA Agent Trained with RL over FHIR EHRs**
  Healthcare QA agent, DAPO, Qwen3-8B, LoRA, verl, vLLM, FHIR – in progress
  - Built a **healthcare QA agent** that answers clinical questions over patient FHIR records by multi-turn ReAct reasoning with **16 deterministic tools**; post-trained **Qwen3-8B** (LoRA) in verl with **DAPO** (clip-higher, zero-variance filtering) on **2,098** FHIR-AgentBench questions at up to **32K-token** contexts, with a verifiable exact-match reward.

University of Virginia, Link Lab (continued) Jun. 2022 – Present
Graduate Research Assistant | *Advisor: Homa Alemzadeh* *Charlottesville, VA*

- **DoC: Learning When to Defer or Commit in Conversational Diagnosis Prediction**
  SFT, GRPO, Qwen3, vLLM, DeepSpeed, W&B – under review
  - Proposed a **self-improving SFT loop** that regenerates turn-level Defer/Commit pseudo-labels with the updated policy from dialogue-level diagnosis labels only, requiring no human-annotated commitment times.
  - Combined Defer-aware SFT initialization with **trajectory-level GRPO** to jointly optimize commitment timeliness, first-commit accuracy, and post-commit stability beyond the supervised policy.
  - Post-trained **Qwen3-1.7B** to outperform Qwen3-4B and Doctor-R1-8B: **78.3% / 52.8%** first-commit accuracy on two real-world dialogue datasets and **43% / 24%** lower post-commit error than the strongest baseline on each.
- **EMSDialog: Synthetic Multi-person EMS Dialogue Generation from Patient Care Reports**
  Multi-LLM agents, EHR-grounded generation, LoRA, vLLM – ACL Findings 2026
  - Built an **EHR-grounded multi-agent generation pipeline** with iterative critique-and-refine, deterministic rule-based checkers, and an LLM style checker to improve clinical factuality, procedural realism, and naturalness.
  - Released **4,414** annotated conversations across **43 diagnoses** with speaker and topic labels; training on EMSDialog-augmented data improved conversational diagnosis accuracy, timeliness, and stability.
- **EMSQA: Expert-Guided Prompting and RAG for EMS Question Answering**
  RAG, FAISS, LoRA, DeepSpeed, vLLM – AAAI 2026 (oral)
  - Created the first EMS multiple-choice QA benchmark (**24.3K questions**, 10 subject areas, 4 certification levels) and a subject-aligned knowledge base of **40K documents and 4M patient care reports**.
  - Developed **Expert-CoT** and **ExpertRAG**: combined, up to **4.67%** accuracy gain over state-of-the-art RAG, and expertise-augmented **32B models passed every EMS certification simulation exam**.
- **DKEC: Domain-Knowledge-Enhanced Multi-Label Diagnosis Prediction**
  Knowledge graphs, UMLS, BERT, GNN, FSDP – EMNLP 2024
  - Built a medical knowledge graph from **3,000+** webpages via chain-of-thought prompting and the UMLS API, yielding **5,000+** normalized entities.
  - Designed **label-wise attention** to fuse the graph with language models, improving micro-F1 by **5%** over state of the art on MIMIC-III.
- **On-Device Multimodal Cognitive Assistant for Emergency Medical Services**
  PyTorch, CUDA, NVIDIA Jetson, ASR, vision-language models – IoTDI 2024, AAAI 2026
  - Cleaned EMS patient care reports with regex normalization and schema mapping; fine-tuned a **15M-parameter EMSTinyBERT** on 4K reports with group-wise training for rare classes.
  - Deployed it on an NVIDIA Jetson at **80.0%** accuracy and **0.31 s** latency; benchmarked zero-shot ASR and VLMs on EgoEMS.

Technical Skills

LLM training & inference: SFT, GRPO / DAPO, LoRA, verl, DeepSpeed, FSDP / DDP, vLLM, Hugging Face, W&B, Slurm
Agents, retrieval & evaluation: multi-agent systems, ReAct tool use, RAG, FAISS, knowledge graphs (UMLS), synthetic data, LLM-as-judge, Gemini / OpenAI APIs
Programming & infra: Python, C/C++, SQL, Java, JavaScript/TypeScript, MATLAB, LaTeX; PyTorch, TensorFlow; Linux, Docker, Git, AWS, Azure, HPC / Slurm, CUDA, NVIDIA Jetson

Honors & Service

Honors: AAAI-26 Student Scholarship (2026); Georgia Tech Merit-Based Scholarship, top 5% (2020); Yangtze Power Scholarship (2018); MCM Meritorious Winner (2018)
Service: Reviewer for ACL, NAACL, ICLR, NeurIPS, AAAI, IJCAI, COLM, ACM TCH; EMNLP 2024 / AAAI 2026 volunteer; TA for two courses; mentored 5 students (four co-authored ACL, AAAI, EMNLP papers)

Selected Publications

- **Ge, X.**, Asgari Mehrabadi, M., & Kuang, J. *FHIR-GRAMS: Graph-Normalized Records and a Multi-Agent System for EHR Question Answering*. **Under review**. Work done as an intern at Samsung Research America.
- **Ge, X.**, Cortez, A., & Alemzadeh, H. *DoC: Learning to Defer or Commit for Timely, Accurate, and Stable Conversational Diagnosis Prediction*. **Under review**.
- **Ge, X.**, Murtaza, S., Cortez, A., & Alemzadeh, H. *EMSDialog: Synthetic Multi-person EMS Dialogue Generation from Electronic Patient Care Reports via Multi-LLM Agents*. **ACL Findings 2026**.
- **Ge, X.**, Murtaza, S., Cortez, A., & Alemzadeh, H. *Expert-Guided Prompting and Retrieval-Augmented Generation for Emergency Medical Service Question Answering*. **AAAI 2026** (**Oral**, 17.6% acceptance).
- Weerasinghe, K., **Ge, X.**, Heick, T., Wijayasingha, L. N., Cortez, A., Satpathy, A., Stankovic, J. A., & Alemzadeh, H. *EgoEMS: A High-Fidelity Multimodal Egocentric Dataset for Cognitive Assistance in Emergency Medical Services*. **AAAI 2026**, Social Impact Track (24.1% acceptance).
- Li, Z., Luo, X., **Ge, X.**, Zhou, L., Lin, X., & Liu, Y. *MMSense: Adapting Vision-based Foundation Model for Multi-task Multi-modal Wireless Sensing*. **IEEE ICC 2026**.
- **Ge, X.**, Satpathy, A., Williams, R. D., Stankovic, J. A., & Alemzadeh, H. *DKEC: Domain Knowledge Enhanced Multi-Label Classification for Diagnosis Prediction*. **EMNLP 2024** (18.4% acceptance).
- Weerasinghe, K., Janapati, S., **Ge, X.**, Kim, S., Iyer, S., Stankovic, J. A., & Alemzadeh, H. *Real-Time Multimodal Cognitive Assistant for Emergency Medical Services*. **IoTDI 2024** (36.7% acceptance).

Patents

- Yu, H., Wang, B., Kong, H., He, W., **Ge, X.**, Yang, W., Li, W., Yang, Z., & Lü, Y. *An action recognition method based on TensorFlow target detection*. **CN111860103B** (granted 2025).
- Yu, H., Kong, H., **Ge, X.**, Wang, B., Wang, Z., Li, W., Yang, Z., & Yu, W. *Automatic gesture recognition system for AD meter understanding capability test*. **CN111652076B** (granted 2024).
- Li, R., **Ge, X.**, Li, Q., Zhao, M., Bao, M., & Ma, J. *Respiratory rate detection method and device, electronic equipment, and storage medium*. **CN113887474B** (granted), SenseTime.
- Li, R., **Ge, X.**, Li, Q., Chen, C., & Ma, J. *Heart rate measurement method and apparatus, and electronic device and storage medium*. **WO2023061042A1** (published 2023), SenseTime.
- **Ge, X.** *Highway electronic information direction board*. **CN209118652U** (granted 2019).